

DRY-RUN GUIDE | pascal_triangle.py

Source: Python\2nd_Belt\pascal_triangle.py

CORE BEHAVIOR

Builds Pascal rows with 1 at the edges and sums of adjacent values above in the interior.

REPRESENTATIVE INPUT

4 rows.

STEP TRACE

Rows: [1]; [1,1]; [1,2,1]; [1,3,3,1].

EXPECTED OUTPUT / RETURN

'1'; '1 1'; '1 2 1'; '1 3 3 1'

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\2nd_Belt\pascal_triangle.py; line numbers refer to the original file. No source was modified.

001 | # Question: Print Pascal's triangle for the given number of rows.

002 | # Add your solution here.

003 |

004 | def print_pascals_triangle(n):

005 | triangle = []

006 | for i in range(n):

007 | # Create a row with 1s

008 | row = [1] * (i + 1)

009 | # Calculate middle values

010 | for j in range(1, i):

011 | row[j] = triangle[i - 1][j - 1] + triangle[i - 1][j]

012 | triangle.append(row)

013 | for row in triangle:

014 | print(" " * (n - len(row)), end="")

015 | print(" ".join(map(str, row)))

016 |

017 |

018 | print_pascals_triangle(5)

019 |

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.